

Average

Name
Date
Course

Average

A number that is an estimate of a **typical** value among a group of numbers.

Ways to Calculate an Average

Arithmetic Mean: The sum of all of the values in a data set divided by the number of values.

Find the arithmetic mean of each data set below.

- ① 23, 36, 31, 38, 29

$$AM = \frac{23 + 36 + 31 + 38 + 29}{5} = \frac{157}{5} = \boxed{31.4}$$

- ② 2, 4, 3, 7, 10, 5, 4

$$AM = \frac{2 + 4 + 3 + 7 + 10 + 5 + 4}{7} = \frac{35}{7} = \boxed{5}$$

Geometric Mean: The n^{th} root of the product of n numbers in a data set.

Find the geometric mean of each data set below.

- ③ 12, 15, 15, 13, $12\frac{1}{3}$, $15\frac{1}{2}$

$$GM = \sqrt[6]{(12)(15)(15)(13)(12\frac{1}{3})(15\frac{1}{2})} = \sqrt[6]{6,709,950} \approx \boxed{13.7}$$

- ④ 8, 4, 7, 7

$$GM = \sqrt[4]{8(4)(7)(7)} = \sqrt[4]{1,568} \approx \boxed{6.29}$$

Median: The middle number in a data set ordered from least to greatest.

Find the median of each data set below.

- ⑤ 62, 50, 49, 47, 59, 60

47, 49, 50, 59, 60, 62 $\boxed{54.5}$

- ⑥ 18, 20, 15, 19, 14

14, 15, 18, 19, 20 $\boxed{18}$

Mode: The number that appears most often in a data set.

Find the mode of each data set below.

- ⑦ 205, 204, 201, 204, 203, 203, 203

$\boxed{203}$

⑧ $\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{3}$

$\frac{1}{3}$

Weighted Average: Calculated by assigning unequal worth to numbers depending on the frequency of their occurrences or their perceived value.

⑨ Bill sold hats, t-shirts, and buttons at the football stadium yesterday. Use the table below to calculate the average price per item of all the things that he sold.

$$AM = \frac{5(16) + 18(15) + 1(124)}{16 + 15 + 124}$$

Item	# sold	Price/item(\$)
Hat	16	5
t-shirt	15	18
button	124	1

$$= \frac{474}{155}$$

$$\approx \$3.058/\text{item}$$

⑩ The table below shows information about all the car wheels sold by a company last year. What is the average diameter of the wheels sold by this company?

$$AM = \frac{20(4) + 15(16) + 18(8) + 17(20)}{4 + 16 + 8 + 20}$$

$$= 16.75 \text{ in}$$

Brand	# sold	Diameter(in)
Rocko	4	20
Jenson	16	15
Altar	8	18
Lobo	20	17

⑪ 98.9% of the carbon atoms on earth have a mass of 12 amu. 1.1% of the carbon atoms have a mass of 13.00335 amu. What is the average mass of the carbon atoms on earth?

$$AM = 0.989(12) + 0.011(13.00335)$$

$$AM = \frac{\# \text{ of } C^{12} \text{ atoms}}{\text{total } \# \text{ atoms}} \text{ mass of } C^{12} + \frac{\# \text{ of } C^{13} \text{ atoms}}{\text{total } \# \text{ atoms}} \text{ mass of } C^{13}$$

$$AM = 12.011 \text{ amu}$$

⑫ 19.78% of the boron atoms on earth have a mass of 10.0129 amu. 80.22% of the boron atoms on earth have a mass of 11.0093 amu. What is the average mass of the boron atoms on earth?

$$AM = 0.1978(10.0129) + 0.8022(11.0093)$$

$$= 10.81 \text{ amu}$$

Average Rate of Change

Average Rate: The **total** change in one quantity divided by the **total** change in a related quantity on a particular interval. **The average rate is a measure of the typical rate during an interval.**

Examples Involving Time

Average Speed: The **total** distance traveled divided by the **total** time passed.

- ⑬ If John drove his car 1,200 miles during his 48 hour trip, what was his average speed **during the whole trip?**

$$\text{Average} = \frac{1200}{48} = \boxed{25 \text{ mi/h}}$$

During the trip, John stopped to sleep and eat. If he drove for a total of 20 hours throughout the entire trip, what was his average speed **during the time that he was driving?**

$$\text{Average} = \frac{1200}{20} = \boxed{60 \text{ mi/h}}$$

- ⑭ On Sunday, Casey spends 3 hours searching for coins and finds 22. On Monday, he spends 5 hours searching and finds 14. He wants to know if collecting coins is worthwhile. To help him decide, he calculates the average number of coins he found per hour **that he searched.** What should this number be?

$$\text{Average Rate} = \frac{22+14}{3+5} = \frac{36}{8} = \boxed{4.5 \text{ coins/h}}$$

Casey had to stay in a hotel for 3 days to search for coins. What is the average number of coins he found per hour **over the course of 3 days?**

$$\text{Average Rate} = \frac{22+14}{3(24)} = \frac{36}{72} = \boxed{\frac{1}{2} \text{ coin/h}}$$

- ⑮ During a race, Natalie ran three yards per second for 200 seconds. She then grew tired and ran the rest of the race at two yards per second for 100 seconds. What was her average speed during the race?

$$\text{Average speed} = \frac{\text{total Distance}}{\text{total time}} = \frac{3(200) + 2(100)}{200+100} = \frac{600+200}{300} = \boxed{2\frac{2}{3} \text{ yds/s}}$$

- ⑯ Nathan went to a ski resort for 5 days. He met 18 people at the resort. What is the average number of people that he met per day **during his entire stay?**

$$\text{Average Rate} = \frac{18}{5} = \boxed{3.6 \text{ people/day}}$$

During his time at the resort, Nathan talked with people for a total of 24 hours. The rest of the time he was alone in his hotel room. What is the average number of people that he met per day **during the time that he talked with people?**

$$\text{Average Rate} = \frac{18}{1} = \boxed{18 \text{ people/day}}$$

- ⑪ Evan rode his bike 8 miles in 30 minutes. He then turned around and rode 6 miles in 25 minutes. What was his average speed?

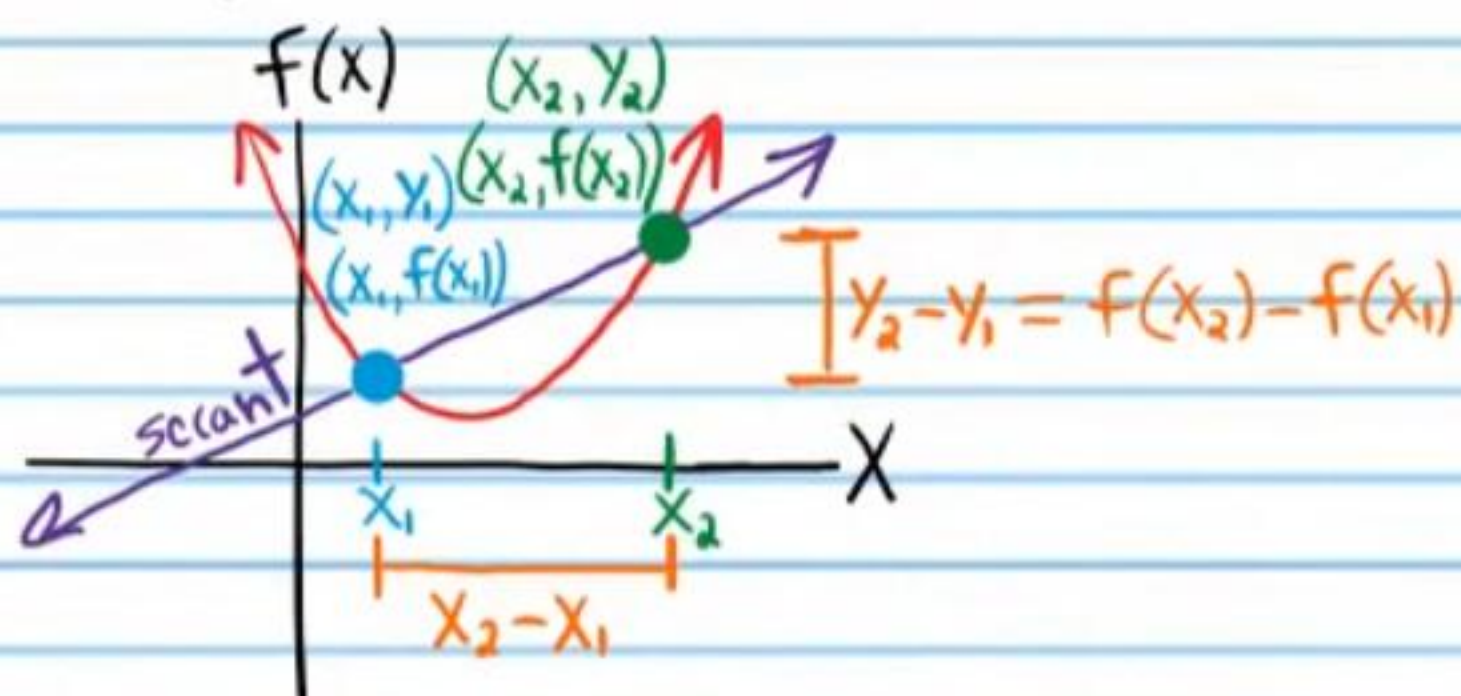
$$\text{Average Speed} = \frac{\text{total distance}}{\text{total time}} = \frac{8+6}{30+25} = \boxed{\frac{14}{55} \text{ mi/min}}$$

- ⑫ A plane flew 350 miles per hour for 3 hours. It then increased its speed to 400 miles per hour and flew for 2 hours. What was the average speed of the plane during the entire distance?

$$\text{Average speed} = \frac{3(350) + 2(400)}{3+2} = \frac{1050+800}{5} = \boxed{370 \text{ mi/h}}$$

Examples Involving Functions

Average Rate of Change of a Function over the Interval x_1 to x_2 .



$$\begin{aligned} \text{Average Rate of Change of the Function} &= \text{Slope of Secant Line} = m = \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} \end{aligned}$$

- ⑬ If $f(x) = -x^2 + 8x - 14$, what is the average rate of change of the function per unit x from $x=1$ to $x=4$?

$$M_{\text{Arc}} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

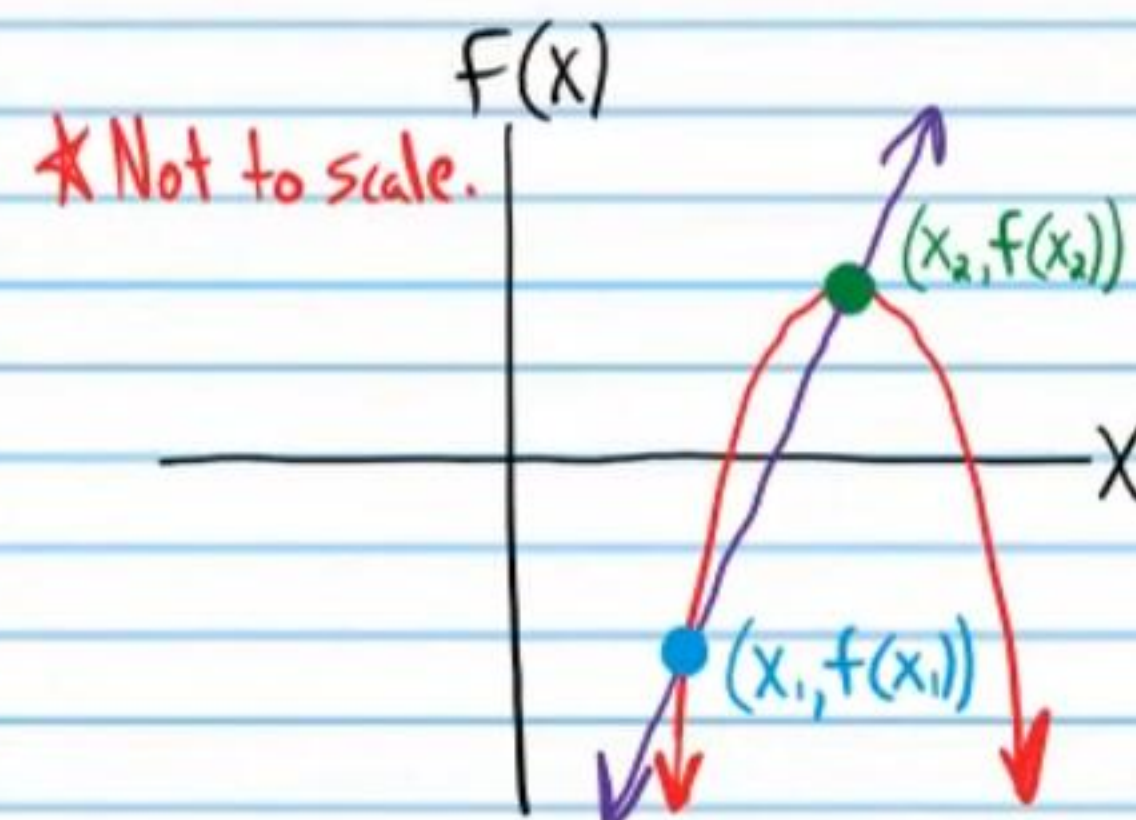
$$= \frac{f(4) - f(1)}{4 - 1}$$

$$= \frac{-4^2 + 8(4) - 14 - (-1^2 + 8(1) - 14)}{3}$$

$$= \frac{-16 + 32 - 14 - (-1 + 8 - 14)}{3}$$

$$= \frac{2 - (-7)}{3}$$

$$= \boxed{3}$$



②0 If $g(x) = -2^x + 8$, what is the average rate of change of the function per unit x from $x = -2$ to $x = 2$?

$$M_{ave} = \frac{y_2 - y_1}{x_2 - x_1}$$

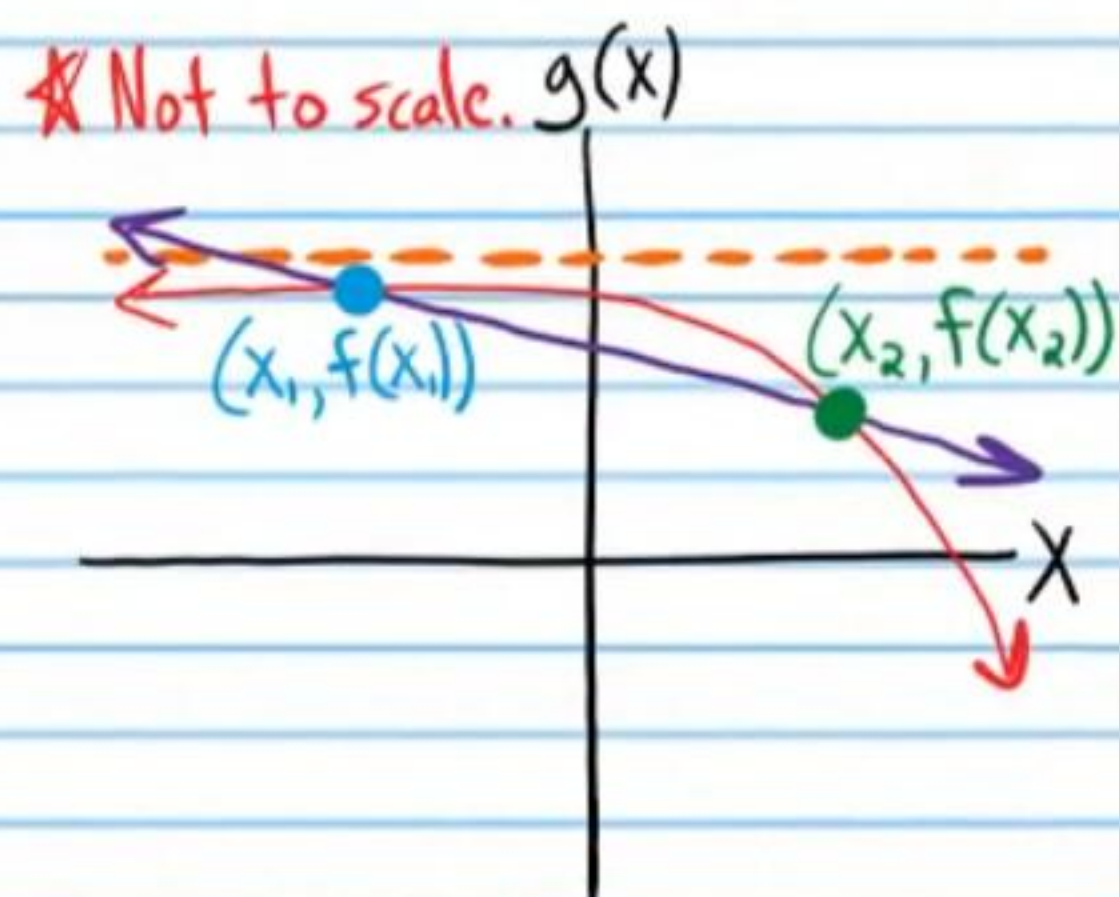
$$= \frac{g(x_2) - g(x_1)}{x_2 - x_1}$$

$$= \frac{g(2) - g(-2)}{2 - (-2)}$$

$$= \frac{-2^2 + 8 - (-2^{-2} + 8)}{2 + 2}$$

$$= \frac{-4 + 8 - (-\frac{1}{2^2} + 8)}{4}$$

$$= \frac{4 - (-\frac{1}{4} + 8)}{4}$$



$$= \frac{4 - 7\frac{3}{4}}{4}$$

$$= \frac{-3\frac{3}{4}}{4}$$

$$= \frac{-\frac{15}{4}}{4}$$

$$= \boxed{-\frac{15}{16}}$$

• ②1 If $h(x) = \frac{1}{3}x^2 + 1$, find the average rate of change of the function per unit x from $x = 0$ to $x = 3$.

$$M_{ave} = \frac{y_2 - y_1}{x_2 - x_1}$$

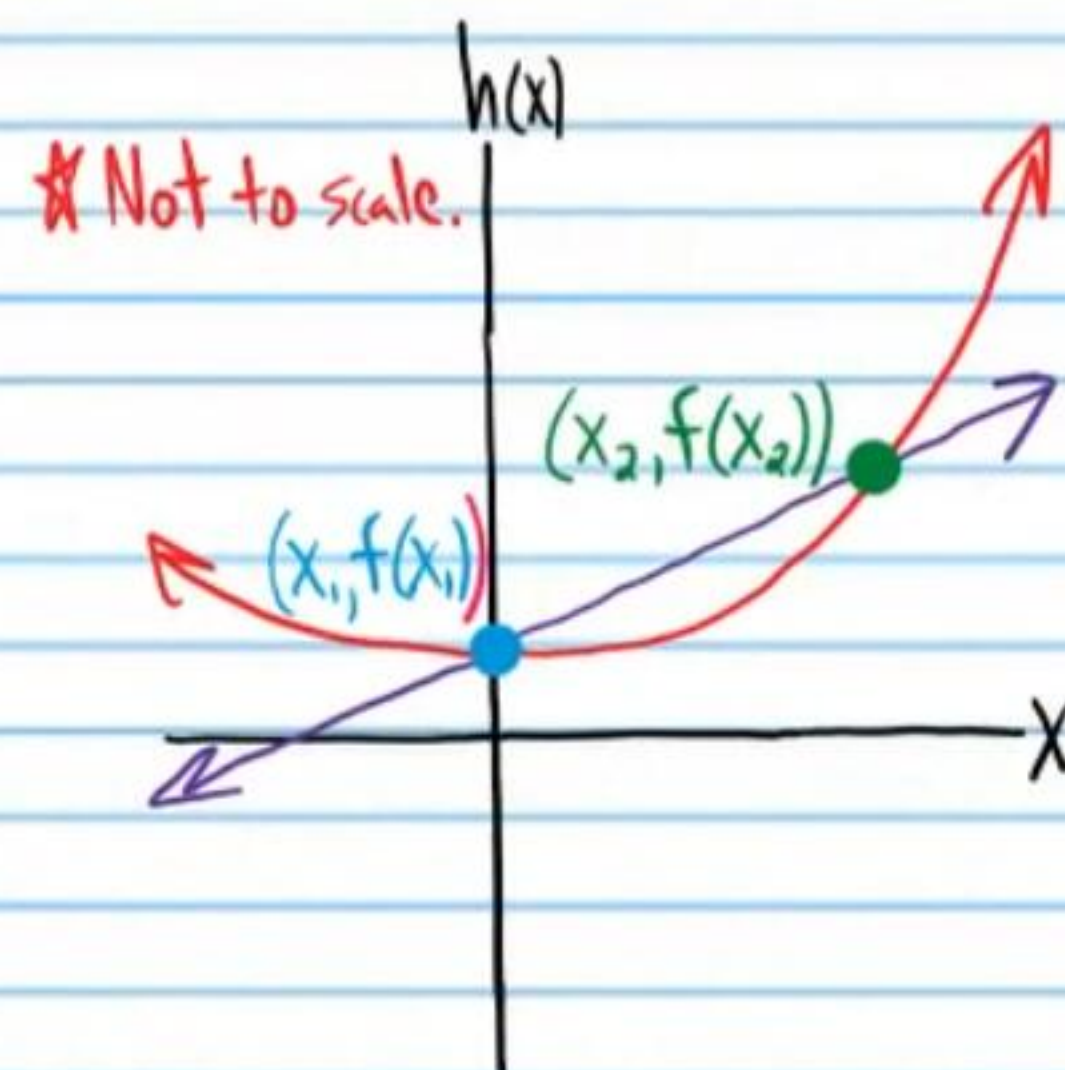
$$= \frac{h(x_2) - h(x_1)}{x_2 - x_1}$$

$$= \frac{h(3) - h(0)}{3 - 0}$$

$$= \frac{\frac{1}{3}(3)^2 + 1 - (\frac{1}{3}(0)^2 + 1)}{3}$$

$$= \frac{\frac{1}{3}(9) + 1 - (0 + 1)}{3}$$

$$= \frac{4 - 1}{3}$$



$$= \boxed{1}$$

② If $f(x) = -2|x+7|$, what is the average rate of change of the function per unit x from $x = -8$ to $x = -5$?

$$M_{ave} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

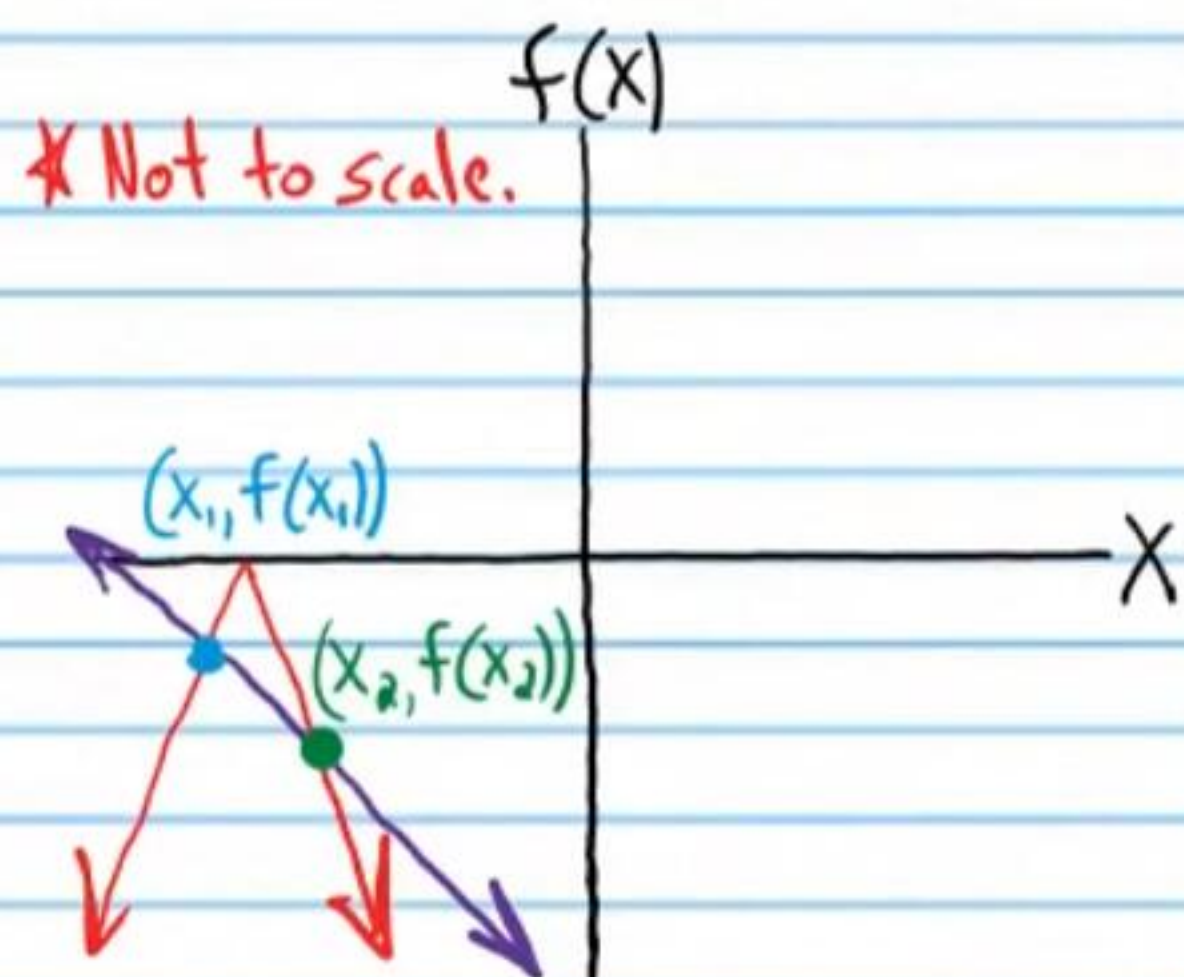
$$= \frac{f(-5) - f(-8)}{-5 - (-8)}$$

$$= \frac{-2|-5+7| - [-2|-8+7|]}{-5+8}$$

$$= \frac{-2|2| - [-2|-1|]}{3}$$

$$= \frac{-2(2) - [-2(1)]}{3}$$

$$= \frac{-4 - (-2)}{3}$$



$$= \frac{-4+2}{3}$$

$$= \boxed{-\frac{2}{3}}$$

More Practice with Function Notation

Given the following functions, find algebraic expressions for the values below.

• ②③ $f(x) = 2x - 3$

i) $f(r) = \boxed{2r - 3}$

ii) $f(2b) = 2(2b) - 3 = \boxed{4b - 3}$

iii) $f(m+5) = 2(m+5) - 3 = 2m + 10 - 3 = \boxed{2m + 7}$

iv) $f(x) + f(x^2) = 2x - 3 + 2x^2 - 3 = \boxed{2x^2 + 2x - 6}$

②④ $g(x) = \frac{1}{\cos|x|-1}$

i) $g(kp) = \boxed{\frac{1}{\cos|kp|-1}}$

ii) $g(3x+5) = \boxed{\frac{1}{\cos|3x+5|-1}}$

iii) $g(t^3+4v) - g(-6t) = \boxed{\frac{1}{\cos|t^3+4v|-1} - \frac{1}{\cos|-6t|-1}}$

$$\textcircled{25} h(x) = \frac{x-5}{x+3} + \log_3 x$$

$$\text{i) } h\left(\frac{p}{2}\right) = \frac{\frac{p}{2}-5}{\frac{p}{2}+3} + \log_3\left(\frac{p}{2}\right) = \boxed{\frac{p-10}{p+6} + \log_3\left(\frac{p}{2}\right)}$$

$$\text{ii) } h(b+5) = \frac{b+5-5}{b+5+3} + \log_3(b+5) = \boxed{\frac{b}{b+8} + \log_3(b+5)}$$

$$\text{iii) } h(a-10) + h(a^6-1)$$

$$\frac{a-10-5}{a-10+3} + \log_3(a-10) + \frac{a^6-1-5}{a^6-1+3} + \log_3(a^6-1)$$

$$\boxed{\frac{a-15}{a-7} + \log_3(a-10) + \frac{a^6-6}{a^6+2} + \log_3(a^6-1)}$$

$$\bullet \textcircled{26} f(x) = \tan x$$

$$\text{i) } f(c) = \boxed{\tan c}$$

$$\text{ii) } f(b^2) = \boxed{\tan b^2}$$

$$\text{iii) } f(m+p+d) = \boxed{\tan(m+p+d)}$$

$$\text{iv) } f\left(\frac{1}{n}\right) + f\left(\frac{n}{3}-1\right) = \boxed{\tan\left(\frac{1}{n}\right) + \tan\left(\frac{n}{3}-1\right)}$$

$$\textcircled{27} g(x) = x^2 - x - 1$$

$$\text{i) } g(\sqrt{r}) = (\sqrt{r})^2 - \sqrt{r} - 1 = \boxed{r - \sqrt{r} - 1}$$

$$\begin{aligned} \text{ii) } g(52-x) &= (52-x)^2 - (52-x) - 1 \\ &= (52-x)^2 - 52 + x - 1 \\ &= \boxed{(52-x)^2 + x - 53} \end{aligned}$$

$$\begin{aligned} \text{iii) } g(\sin b - \cot b) &= (\sin b - \cot b)^2 - (\sin b - \cot b) - 1 \\ &= \boxed{(\sin b - \cot b)^2 - \sin b + \cot b - 1} \end{aligned}$$

$$\begin{aligned} \text{iv) } g(|x|) - g(x+1) &= |x|^2 - |x| - 1 - [(x+1)^2 - (x+1) - 1] \\ &= |x|^2 - |x| - 1 - (x+1)^2 + x+1 + 1 \\ &= \boxed{|x|^2 - |x| - (x+1)^2 + x+1} \end{aligned}$$

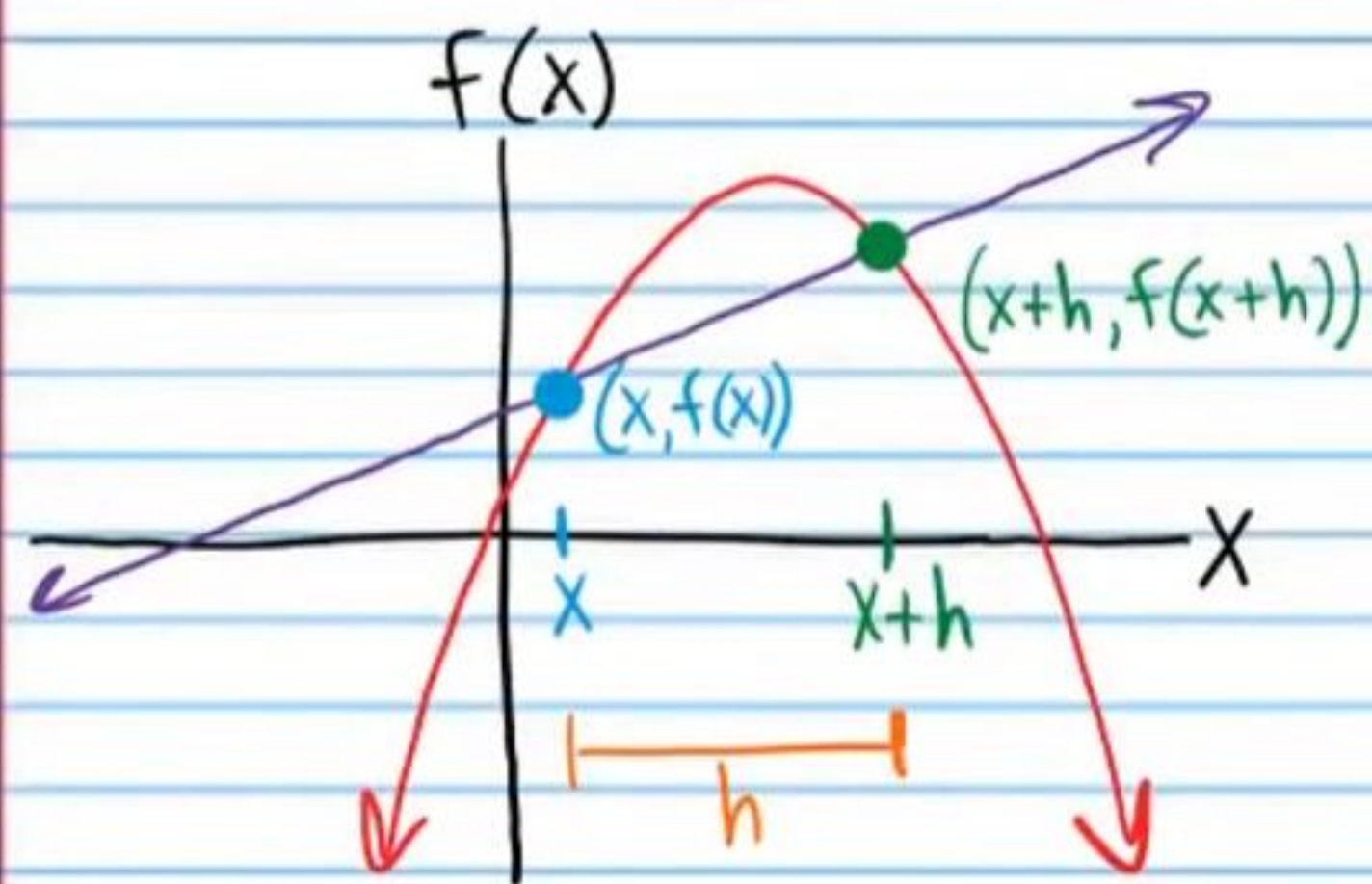
$$\textcircled{28} h(x) = \frac{1+2x}{1+3x}$$

$$\text{i) } h(-6) = \frac{1+2(-6)}{1+3(-6)} = \frac{1-12}{1-18} = \frac{-11}{-17} = \boxed{\frac{11}{17}}$$

$$\text{ii) } h\left(-\frac{2x}{3}\right) = \frac{1+2\left(-\frac{2x}{3}\right)}{1+3\left(-\frac{2x}{3}\right)} = \frac{1-\frac{4x}{3}}{1-2x} = \boxed{\frac{3-4x}{3-6x}}$$

$$\text{iii) } h(1-x) = \frac{1+2(1-x)}{1+3(1-x)} = \frac{1+2-2x}{1+3-3x} = \boxed{\frac{3-2x}{4-3x}}$$

The Difference Quotient



$h = \text{change in } x.$

The Difference Quotient = Average Rate of Change of a Function

= slope of the secant line

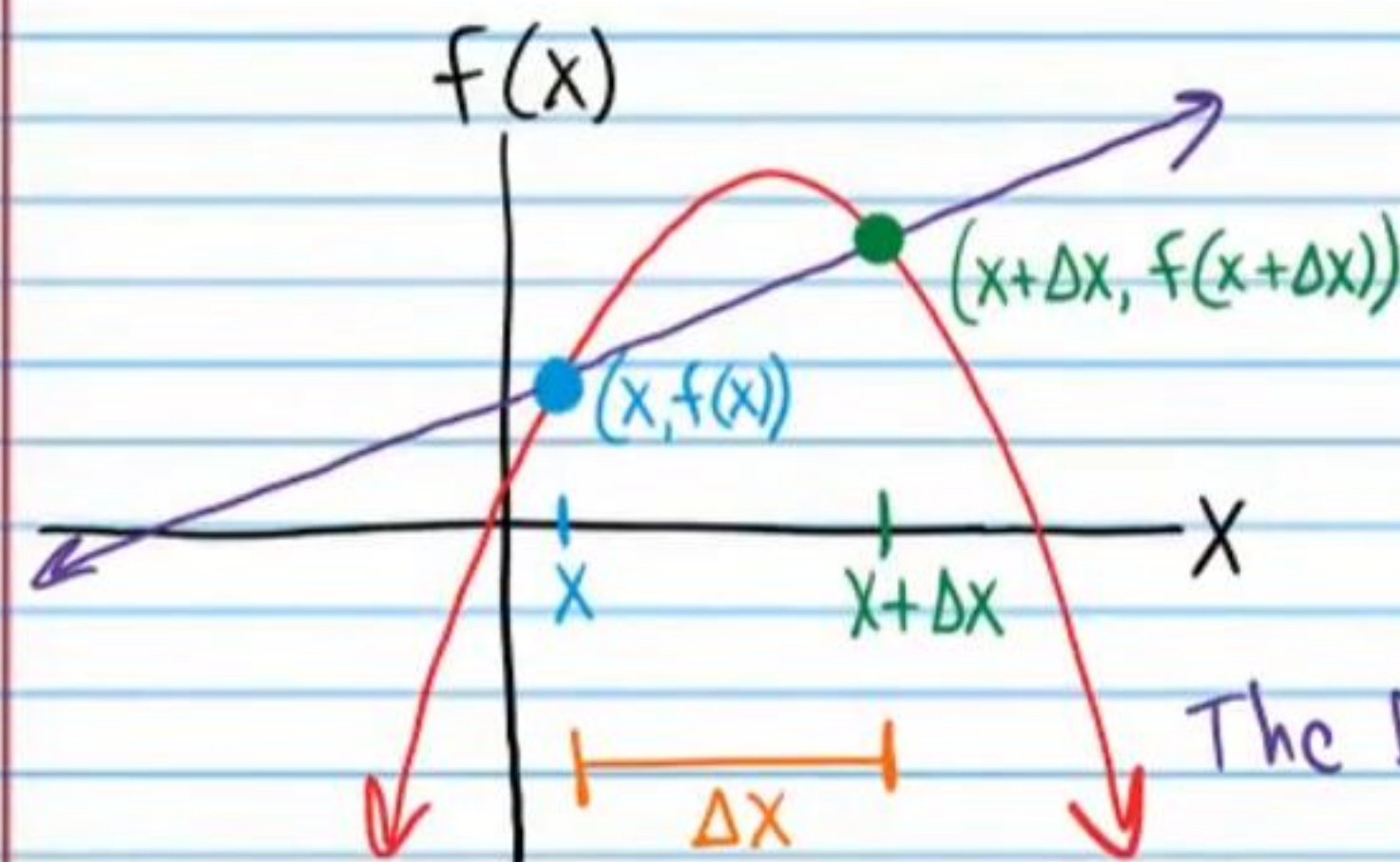
= m

$$= \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{f(x+h) - f(x)}{x+h - x}$$

$$= \frac{f(x+h) - f(x)}{h}$$

Alternate Notation ($\Delta x = h$)



$\Delta x = \text{change in } x.$

The Difference Quotient

$$\frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Expressing the Difference Quotient

For each function, find an algebraic expression representing the difference quotient (use "h" for change in "x"). It is not necessary to simplify each expression.

• (29) $f(x) = x^3 + x - 4$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{(x+h)^3 + x+h - 4 - (x^3 + x - 4)}{h}$$

(30) $F(x) = 2x^2 - 1$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{2(x+h)^2 - 1 - (2x^2 - 1)}{h}$$

$$(31) f(x) = \sin x + 1$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\sin(x+h) + 1 - (\sin x + 1)}{h}$$

$$(32) f(x) = \frac{1}{x-3} + \log_2 x$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\frac{1}{x+h-3} + \log_2(x+h) - \left(\frac{1}{x-3} + \log_2 x\right)}{h}$$

$$(33) f(x) = |x| + 3^x$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{|x+h| + 3^{x+h} - (|x| + 3^x)}{h}$$

$$(34) f(x) = x + 10$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{x+h+10 - (x+10)}{h}$$

$$(35) f(x) = -\frac{2}{x} + \cot x^3 - 4x$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{-\frac{2}{x+h} + \cot(x+h)^3 - 4(x+h) - \left(-\frac{2}{x} + \cot^3 x - 4x\right)}{h}$$

$$(36) f(x) = x$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{x+h - x}{h}$$

$$(37) f(x) = \frac{x+1}{x-2}$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\frac{x+h+1}{x+h-2} - \frac{x+1}{x-2}}{h}$$

$$\bullet (38) f(x) = x^2$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h}$$

$$(39) f(x) = x - 9x^3$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{x+h - 9(x+h)^3 - (x - 9x^3)}{h}$$

$$(40) f(x) = \cos x$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\cos(x+h) - \cos x}{h}$$

$$(41) f(x) = 1 - \log_3 x$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{1 - \log_3(x+h) - (1 - \log_3 x)}{h}$$

$$(42) f(x) = \frac{1}{x}$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$$

$$(43) f(x) = \frac{\tan x}{\log_4 x}$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\frac{\tan(x+h)}{\log_4(x+h)} - \frac{\tan x}{\log_4 x}}{h}$$

$$(44) f(x) = x(\sec x - 3)$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{(x+h)(\sec(x+h) - 3) - (x(\sec x - 3))}{h}$$

$$(45) f(x) = 2^{\sec x}$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{2^{\sec(x+h)} - 2^{\sec x}}{h}$$

$$(46) f(x) = \frac{10^x}{3}$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\frac{10^{x+h}}{3} - \frac{10^x}{3}}{h}$$

$$(47) f(x) = \frac{x+10}{3-2x}$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\frac{x+h+10}{3-2(x+h)} - \frac{x+10}{3-2x}}{h}$$

$$(48) f(x) = \frac{x}{\sin x} + \frac{2}{\cos x}$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{\frac{x+h}{\sin(x+h)} + \frac{2}{\cos(x+h)} - \left(\frac{x}{\sin x} + \frac{2}{\cos x} \right)}{h}$$